

Statistical evaluation of PM₁₀ and distribution of PM₁, PM_{2.5}, and PM₁₀ in ambient air due to extreme fireworks episodes (Deepawali festivals) in megacity Delhi

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Abstract Temporal variation of PM₁₀ using 2-year data (January, 2007–December, 2008) of Delhi is presented. PM₁₀ varied from 42 to 200 $\mu\text{g m}^{-3}$ over January to December, with an average $114.1 \pm 81.1 \mu\text{g m}^{-3}$. They are comparable with the data collected by Central Pollution Control Board (National Agency which monitors data over the entire country in India) and are lower than National Ambient Air Quality (NAAQ) standard during monsoon, close to NAAQ during summer but higher in winter. Among CO, NO₂, SO₂, rainfall, temperature, and wind speed, PM₁₀ shows good correlation with CO. Also, PM₁₀, PM_{2.5}, and PM₁ levels on Deepawali days when fireworks were displayed are presented. In these festive days, PM₁₀, PM_{2.5}, and PM₁ levels were 723, 588, and 536 $\mu\text{g m}^{-3}$ in 2007 and 501, 389, and 346 $\mu\text{g m}^{-3}$ in 2008. PM₁₀, PM_{2.5}, and PM₁ levels in 2008 were 1.5 times lower than those in 2007 probably due to higher mixing height (446 m), temperature (23.8°C), and winds (0.36 ms^{-1}).

Keywords Hazardous particles · Fireworks · Mixing height · Air quality · Vehicular pollution

1 Introduction

Ambient aerosols with mass median aerodynamic diameter less than 10 μm (PM₁₀), 2.5 μm (PM_{2.5}), and 1 μm (PM₁) are of interest owing to their effects on human health, visibility, and climate (Watson 2002; Pope and Dockery 2006; IPCC 2007; Pope et al. 1995, 2004; Chate 2011). Increasing particulate matter has already adversely affected human health in megacities (>10 million people; Madronich 2006). In megacities, main

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sources of PM_{10} , $PM_{2.5}$, and PM_1 are the combustion of fossil fuels from automobiles, construction equipments (mobile sources), furnaces, and power plants (stationary sources). Fine particles ($PM_{2.5}$ and PM_1) are emitted by combustion, while coarse particles (PM_{10}) are blown into the ambient air by mechanical processes (Faiz et al. 1992; Madronich 2006). PM_{10} , CO, NO_2 , O_3 , SO_2 , and volatile organic compounds are periodically monitored and regulated as they have complex, additive, synergistic, or sequential biological effects (Morawska et al. 2002; Maynard and Kuempel 2005). For Instance, PM_{10} and CO, NO, and NO_2 have been assessed over 6 weeks in the Tuhobic road tunnel (2,140 m), Croatia (Ivan et al. 2005). However, regression analysis of PM_{10} with trace gases and meteorological parameters using relatively large data is lacking for megacities and sparse in the literature. The present study thus assumes significance and forms the subject matter in this first part.

Firecrackers are made up of variety of chemicals (potassium nitrates, potassium chlorate, potassium per-chlorate, charcoal, sulfur, manganese, sodium oxalate, aluminum and iron dust powder, strontium nitrate, barium nitrate, etc.; McLain 1980). On combustion, they release numerous hazardous pollutants (sulfur dioxide, carbon dioxide, carbon monoxide, suspended particles, aluminum, manganese cadmium, etc.) that affect human health (Hirai et al. 2000; Ravindra et al. 2001). Bach et al. (1975) reported an increase in total suspended particulate matter (TSPM) on an average of 300% at 14 locations and by 700% in the lung penetrating size ranges at one location due to fireworks on New Year's Eve. Increase in particle number is witnessed in the accumulation mode range (>100 nm) during the Millennium fireworks in Leipzig, (Germany; Wehner et al. 2000). Liu et al. (1997) reported the chemical composition and particle size of typical firework mixtures. Rise in $PM_{2.5}$ up to 6 and PM_{10} up to 4 times on a lantern day (fireworks) in Beijing (China) found relative to those over normal days (Wang et al. 2007).

In India, during Deepawali, fireworks on huge scale especially in urban areas add large amount of anthropogenic pollutants into local environment. For example, Ba, K, Al, and Sr levels rose up to 1,091, 25, 18, and 15 times higher (Kulshrestha et al. 2004), black carbon increased by a factor of 3 (Babu and Moorthy 2001), and SO_2 , NO_2 , PM_{10} , TSP by 2–10 times (Ravindra et al. 2003) during Deepawali. Several interesting results are reported in literature related to Deepawali fireworks over different parts of India (Agarwal et al. 2006; Barman et al. 2008; Kakoli and Gupta 2007; Kulshrestha et al. 2004; Singh et al. 2003; Singh et al. 2008). Babu and Moorthy (2001) observed a large increase in carbon particles after Deepawali festival in Thiruvananthapuram. Three fold increases in PM_{10} and TSPM were attributed to fireworks at Hisar, India (Ravindra et al. 2003). Kulshrestha et al. (2004) documented high levels of different trace elements in air at Hyderabad, (India). For the first time, Attri et al. (2001) found O_3 formation by sparklers during Deepawali nights at Delhi. To the authors' knowledge, very few measurements of PM_{10} , $PM_{2.5}$, and PM_1 have been made during firework episodes of Deepawali, over the Indian region.

In this paper, the results are presented in the following order (i) Temporal variation of PM_{10} , (ii) regression analysis of PM_{10} with CO, NO_2 , O_3 , and SO_2 and meteorological parameters, and (iii) Variations in PM_{10} , $PM_{2.5}$, and PM_1 levels two extreme firework episodes (Deepawali-2007 and 2008). At the end, PM_1 variations in the ambient air during these firework display episodes at Delhi are discussed in terms of health hazard.

2 Observational Site

Delhi ($28^{\circ}21'17''$ – $28^{\circ}53'$ latitude and $76^{\circ}20'37''$ – $77^{\circ}20'37''$ longitude) is situated 160 km South of the Himalayas, at altitudes from 213.3 m to 305.4 m above mean sea level. It is

bounded by the Thar-Desert of Rajasthan in the West and plains of central India in the South. The megacity extends over 1,483 km² with the present population of ~18 millions relative to ~3.5 millions in 1970. The entire northern part of India, especially Indo-Gangetic Plain, experiences a thick foggy weather during winter coupled with lower mixing heights. Consequently, pollutants do not disperse or mix freely with the upper boundary layer and thus rendered visibility poor. The temperature decreases rapidly toward the end of October, and continental air masses rich in pollutants pass over Delhi during winter. Winds are predominantly Westerly or North-Westerly during winter and Easterly and South-Easterly in monsoon. Besides numerous domestic coal burning units and three thermal power plants with the combined capacity of 1,087 MW operating in different parts of Delhi, vehicular and industrial emissions are major sources of air pollutants.

Daily monitored PM₁₀, CO, NO₂, and SO₂ data at Income Tax Office in central part of Delhi are retrieved from CPCB web site (<http://www.cpcb.nic.in>) for January, 2007 to December, 2008. A thermal power plant (747 MW) is located within 500 m in the southeast direction of this site. Rainfall, temperature, RH, and wind speed data are collected from the India Meteorological Department (<http://www.imd.gov.in>).

Measurements of PM₁₀, PM_{2.5}, and PM₁ were carried out with a GRIMM aerosol spectrometer (Model 1.108: OPC, GRIMM Inc.) at about 15 m above the ground level, on the rooftop of Indian Institute of Tropical Meteorology (IITM) building located in the central part of Delhi during Deepawali of 2007 and of 2008. This site primarily is a residential area away from the large pollutant-sources. GRIMM aerosol spectrometer is specifically designed for monitoring PM₁₀, PM_{2.5}, and PM₁ aerosols. The technology enables to make precise cut off diameters for all the three PM sizes. This system allows collection of all three PM fractions simultaneously without any necessity of changing sampling heads. The particle analyzer was operated continuously a day before, on, and after Deepawali festivals for 2 days. A constant flow rate 1.2 liter/min was maintained throughout measurements.

3 Results and discussion

3.1 Temporal variation of PM₁₀ concentrations

Time-series of PM₁₀, and of temperature, wind, rainfall, and relative humidity from January, 2007 to December, 2008 are plotted in Figs. 1 and 2, respectively. Monthly mean PM₁₀ levels varied between 42 and 200 µg/m³ with an annual arithmetic mean 114.1 ± 81.7 µg m⁻³. PM₁₀ concentrations during monsoon were well below the National Ambient Air Quality (NAAQ) (60 µg m⁻³) standard but were much above the NAAQ standard during winter. However, they were close to NAAQ standard during summer. The mean PM₁₀ during November was highest (374 µg m⁻³ in 2007 and 220 µg m⁻³ in 2008), while minimum concentration was registered during August (31 µg m⁻³ in 2007 and 42 µg m⁻³ in 2008). The annual mean PM₁₀ (114.1 µg m⁻³) was lower than the previously reported (171 µg m⁻³) by Khillare et al. (2008), during 2004 for Delhi, which can be attributed to introduction of metro-rail and use of CNG for public transport in Delhi. Considering the 24 h NAAQ standard (100 µg m⁻³), 58% exceedances in 2007 and 50% exceedances in 2008 were registered. While mean PM₁₀ of 221 µg m⁻³ in winter is attributed to high relative humidity, low winds, and temperatures, 90 µg m⁻³ in summer are due to strong winds and higher temperatures.

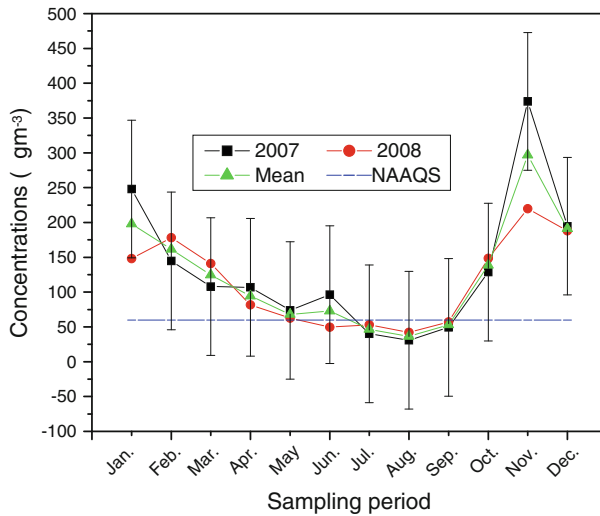


Fig. 1 Monthly arithmetic mean variation of PM_{10} concentrations during 2007 and 2008 along with the values of NAAQS Standard

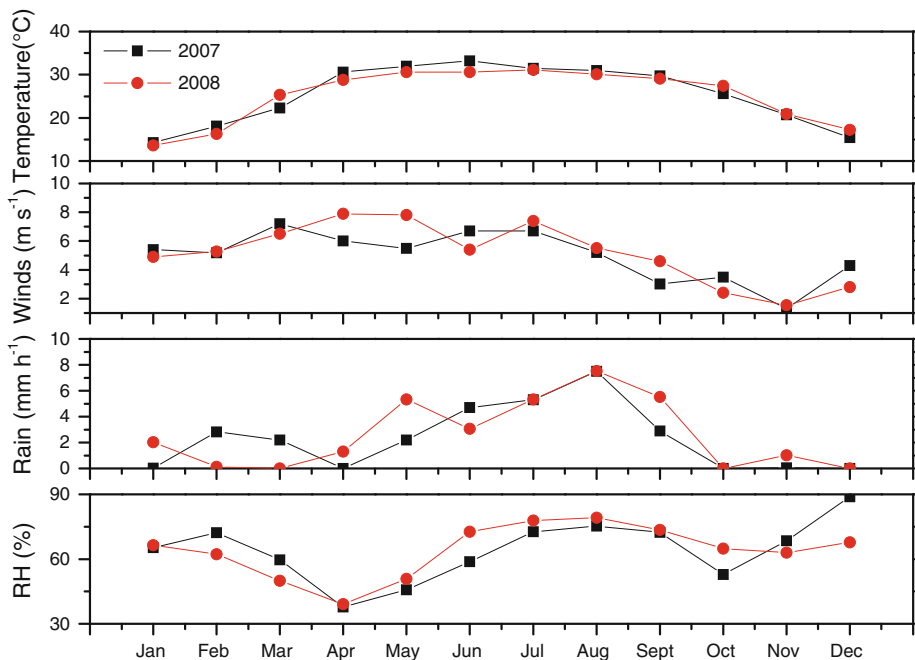


Fig. 2 Monthly mean of RH, rainfall, winds, and temperature during 2007 and 2008

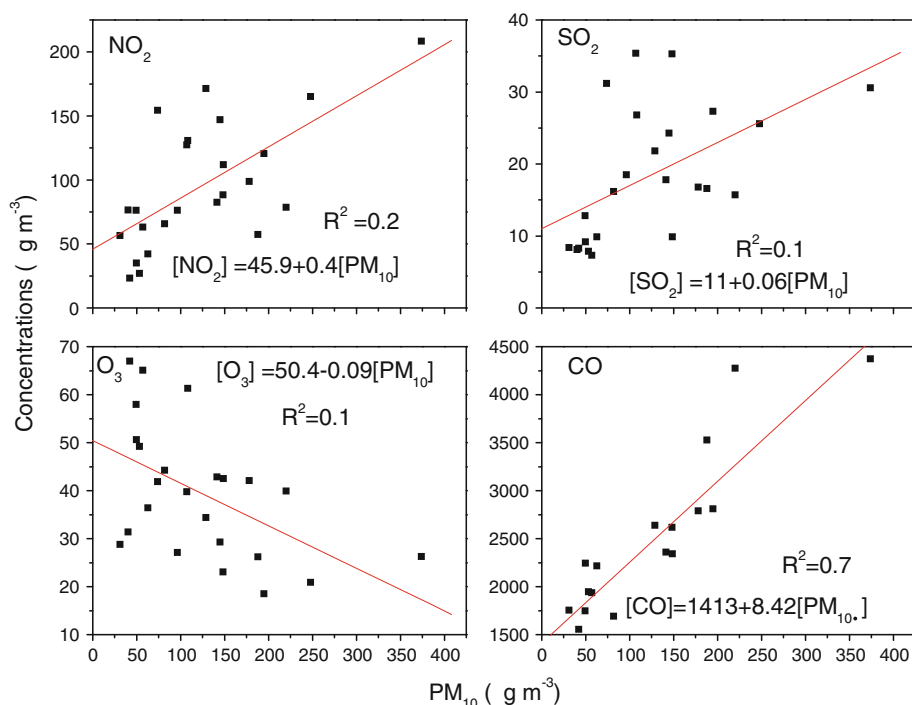
3.2 Relationship between of PM_{10} and gaseous pollutants

Pearson correlation analyses were made for PM_{10} with CO , NO_2 , O_3 , and SO_2 and are tabulated in Table 1. Linear regression analyses in Fig. 3 show better correlation of PM_{10}

Table 1 Pearson correlation coefficients between PM₁₀ concentrations and concentrations of gaseous pollutants

	CO	SO ₂	O ₃	NO ₂
PM ₁₀	0.90**	0.53**	−0.52**	0.66**

** Correlations significant at the 99% confidence level


Fig. 3 Scatter plots with linear regression between PM₁₀ and gaseous pollutants

with CO than with NO₂ and SO₂. Least-square fit relationships indicate that the rising trend of PM₁₀ is better related to the increase in CO compared to NO₂ and SO₂. Ivan et al. (2005) have observed highest correlation of PM₁₀ with NO₂, while among the trace gases (CO, NO₂, O₃, and SO₂), NO and CO are better correlated at the Tuhobc road tunnel (2,140 m), Croatia. Mean CO, NO₂, O₃, and SO₂ concentrations were 2,400, 95, 40, and 19 µg/m³, respectively, at Delhi. Among gaseous pollutants, NO₂ and CO levels often exceeded the permissible limit of NAAQ (viz., 80 and 2,000 µg/m³ annual standard for NO₂ and CO). Vehicular pollution is the main contributor (~72%) of NO₂ levels in Delhi (Jain and Khare 2008) apart from the fuel combustion by both large-scale and small-scale industries. The probable cause of O₃ is expected to be photochemistry but yet to be confirmed as an atmospheric process (Attri et al. 2001; Ali et al. 2009). Of the large-scale industries, thermal power plants, namely, Rajghat, Indra Prastha, and Badarpur, of total generation capacity, 1,087 MW, have been most prominent sources of air pollutions in Delhi (CPCB 2001). Additionally, cumulative effects of population growth, industrialization, and frequent operation of large number of Diesel Generator-Sets (by various commercial and

industrial establishments) are responsible for significant air-pollution in Delhi. Also, mean NO_2 levels were exceeding annual mean standard set by World Health Organization ($40 \mu\text{g}/\text{m}^3$). Regression analysis of PM_{10} with NO_2 (as an indicator for road traffic emissions) revealed a contribution of vehicular emissions. NO_2 and PM_{10} concentrations were high during winter and low during monsoon. High pollution levels in winter were due to lower dispersion of pollutants on account of weak winds. On the other hand, high particulate pollution levels in the summer were due to entrainment of dust—owing to dust storms and high winds. Also, North-westerly winds bring additional burden to particulates from the neighboring desert from the state of Rajasthan (India). Thus, there is huge influx of dust/particulates from Rajasthan, which surrounds the Delhi from the North to West and the South to West. On one hand higher winds lead to greater influx of particulates while on the other leads to decreased gaseous pollution due to their higher dispersion and mixing capacity.

3.3 Regression analysis between PM_{10} and meteorological parameters

Linear regression analysis of PM_{10} with temperature, wind speed, RH, and rainfall is presented in Fig. 4. PM_{10} decreases with increase of temperature, wind speed, and rainfall but varies slightly with RH. It is inversely correlated with temperature, wind speed, and rainfall with coefficients as -0.53 , -0.35 , and -0.5 , respectively. While high winds disperse PM_{10} , low temperature, RH, and wind speed favors build-up of pollution vehicular. Increased levels of PM_{10} in Netherland during winter are found to occur during dry

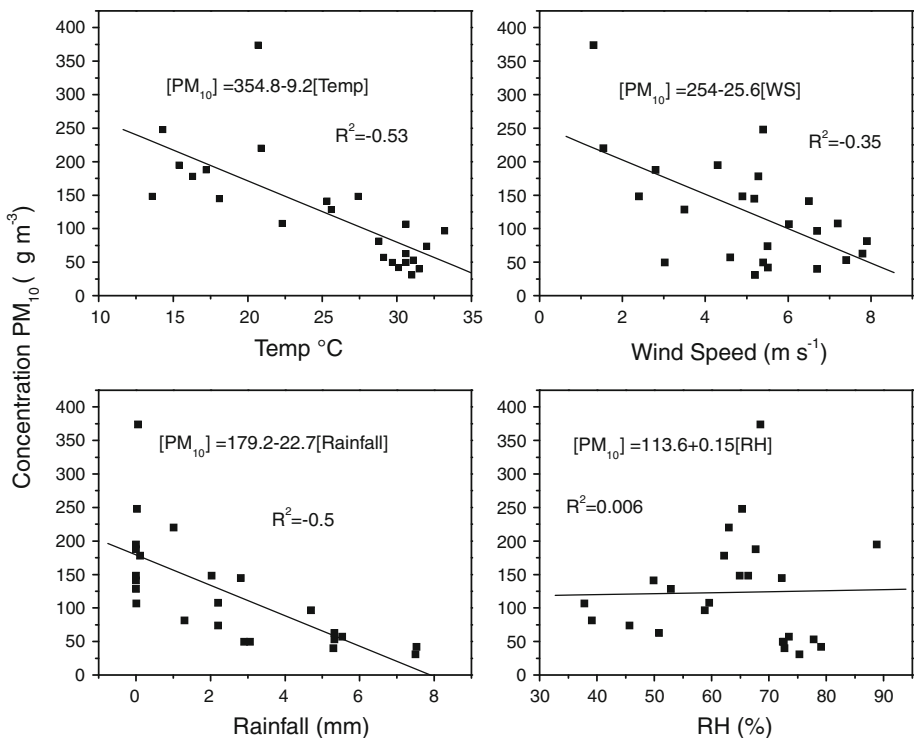


Fig. 4 Linear regression analysis between PM_{10} and meteorological parameters

weather with low temperatures (Vander Wal and Janssen 2000), which is in agreement with higher PM_{10} levels at Delhi during winter.

3.4 Variation of PM_{10} , $PM_{2.5}$, and PM_1 concentrations during Deepawali festivals

During Deepawali celebration, people display fireworks on a large scale resulting in noise, release of toxic gases, dust, and solid waste into the atmosphere, which affects adversely the public in general and asthmatic patients in particular. Measurements of PM_{10} , $PM_{2.5}$, and PM_1 were carried out at the Institute Building (Delhi) during 8–11 November, 2007 (Deepawali day was on 9th November) and during 27–30 October, 2008 (Deepawali day was on 28th October). Round-the-clock PM_{10} , $PM_{2.5}$, and PM_1 observations were taken a day before, on, and after the Deepawali day for 2 days and plotted in Figs. 5a, b. The mean of PM_{10} , $PM_{2.5}$, and PM_1 concentrations were 615, 495, and 444 $\mu\text{g}/\text{m}^3$. It is worth to note that PM_{10} , $PM_{2.5}$, and PM_1 concentrations peaked at mid-night on festival day (9th November) and at 0500 h on the following day (10th November) for Deepawali of 2007. Average of PM_{10} , $PM_{2.5}$, and PM_1 concentrations over festive days were 723, 588, and 536 $\mu\text{g}/\text{m}^3$ in 2007 and 501, 389, and 346 $\mu\text{g}/\text{m}^3$ in 2008, respectively. Lower PM_{10} , $PM_{2.5}$, and PM_1 levels registered on festive days in 2008 (October 28) compared to that in 2007 (November 9) can be attributed to increased mixing height (446 m), temperature (23.8°C), and wind (0.36 ms^{-1}) in 2008 compared to those in 2007 (301 m, 21.1°C and 0.10 ms^{-1}).

Central Pollution Control Board (CPCB) monitors respirable suspended particulate matter (PM_{10}) at 5 residential and 2 industrial sites at Delhi using Fine Particulate Sampler (APM 541; Envirotech Instruments Pvt. Ltd, design approved by USEPA), maintaining flow rate 1 m^3/h during observational period. A 47-mm quartz micro-filter (Whatman, UK) was used for sampling. Ambient aerosol mass concentrations at various locations during Deepawali days in the years 2004, 2005, 2006, 2007, and 2008 are summarized in Table 2 (<http://www.cpcb.nic.in/>). Average aerosol levels at these 7 locations over Delhi (5 Residential and 2 Industrial) sequentially in the above years were in the range 607–1797, 337–552, 265–440, 610–1,294, and 578–931 $\mu\text{g}/\text{m}^3$, respectively. Average PM_{10} concentrations monitored by the authors at IITM, Delhi (residential site) using GRIMM analyzer system was 723 $\mu\text{g}/\text{m}^3$ over festive days in 2007, which is also in the range (610–1,294 $\mu\text{g}/\text{m}^3$) reported by CPCB, but average of PM_{10} (501 $\mu\text{g}/\text{m}^3$) in 2008 was lower than that of the range reported by CPCB (578–931 $\mu\text{g}/\text{m}^3$). However, PM_{10} concentrations during Deepawali festivals reported either by CPCB (2004, 2005, 2006, 2007, and 2008) or in the present study (2007 and 2008) were much higher than the ambient air-quality standards for residential (100 $\mu\text{g}/\text{m}^3$) and for industrial (150 $\mu\text{g}/\text{m}^3$) sites.

An adult inhales 10–20 m^3 (12–24 kg) of air in a day by respiratory processes (Hind 1999) considering diurnal variations of initial PM_1 concentrations, and assuming mean deposition fraction as 0.63 (Chate 2011), a rough estimation of diurnal variations of PM_1 particle deposition within respiratory system on Deepawali days of 2007 and 2008 is illustrated in Fig. 6. Rough estimates from these initial and deposited concentrations of PM_1 with sampling hours (see Fig. 6) show 37% deposit within respiratory system for Deepawali days of 2007 and 2008. However, deposition of inhaled PM_1 particles in the various regions of the respiratory system is a complex function of the impaction, sedimentation, diffusion, interception, and electrostatic deposition mechanism, discussion of which is beyond the scope of this work.

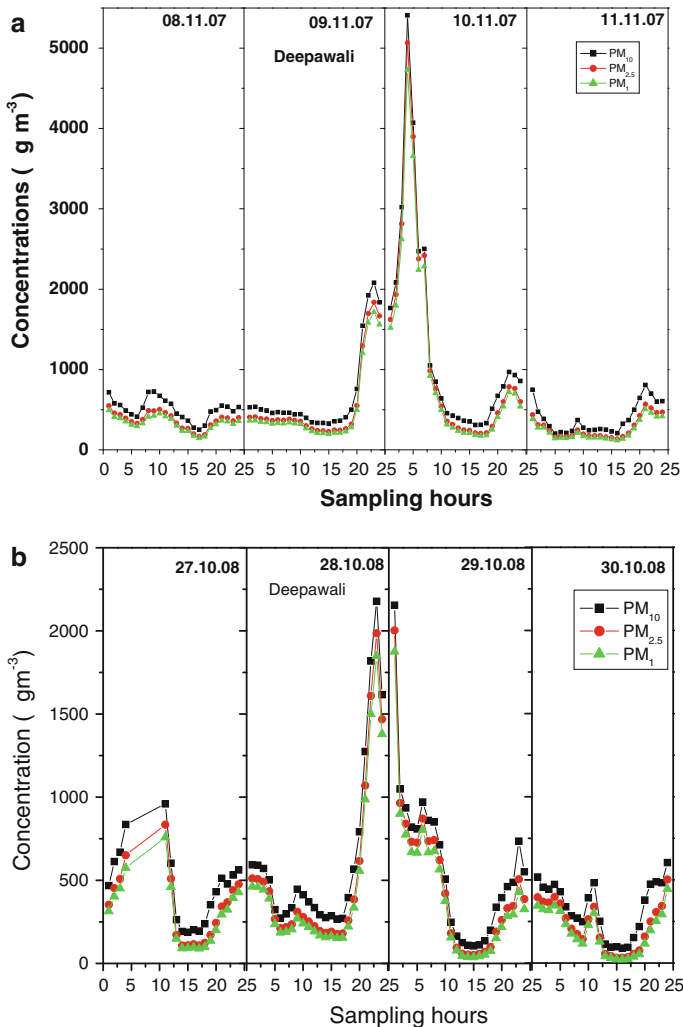


Fig. 5 **a** Diurnal variations of PM_{10} , $PM_{2.5}$, and PM_1 from 8–11.11.2007 during the Deepawali festival. **b** Diurnal variations of PM_{10} , $PM_{2.5}$, and PM_1 from 27 to 30.10.2008 during the Deepawali festival

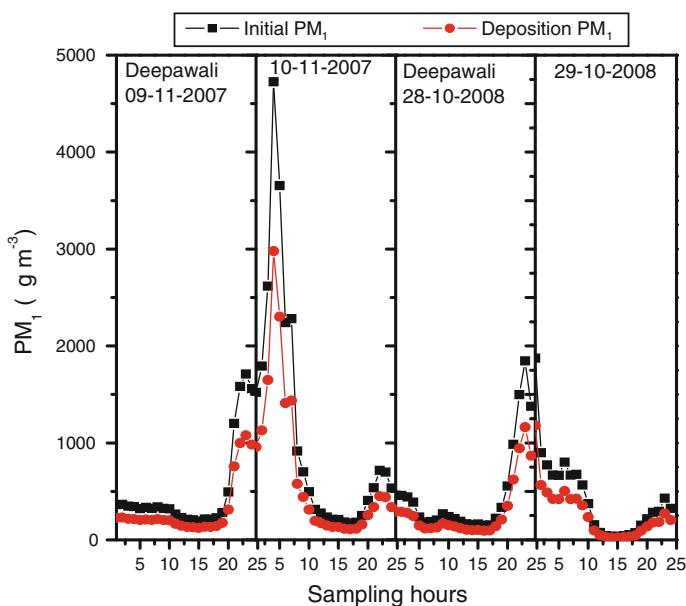
4 Conclusions

Long-term data (January, 2007 to December 2008) revealed that PM_{10} concentrations at Delhi varied between 42 and 200 $\mu g/m^3$ with an annual mean of $114.1 \pm 81.7\ \mu g\ m^{-3}$. PM_{10} ($221\ \mu g\ m^{-3}$) is 4 times higher than NAAQ Standard ($PM_{10} = 60\ \mu g/m^3$) during winter, one and half times higher ($90\ \mu g/m^3$) during summer and lower during monsoon ($<60\ \mu g/m^3$). These seasonal patterns in PM_{10} are due to emissions from fireworks during Deepawali (winter), windblown dust in summer and effective aerosol scavenging by extensive rain during monsoon. Inverse correlations of PM_{10} with temperature (-0.53), wind speed (-0.35), and rainfall (-0.5) support the seasonal variation patterns.

Table 2 Ambient aerosol levels at various locations during Deepawali 2004, 2005, 2006, 2007, and 2008 [Residential (R); Industrial (I)], CPCB (<http://www.cpcb.nic.in>) Range for Delhi (Present study)

Ambient aerosols mass ($\mu\text{g m}^{-3}$)					
Year	2004	2005	2006	2007	2008
B.S.Z Marg (ITO) (R)	896	378	304	670	578
Ashok Vihar (R)	1,222	552	388	1,294	712
Sirifort (R)	936	489	440	1,139	901
Janakpuri (R)	607	422	426	1,068	931
Nizamuddin (R)	1,161	337	265	1,012	703
Shahazada Bagh (I)	827	520	300	966	719
Shahdara (I)	1,797	407	395	610	708
Range for Delhi	607–1,797	337–552	265–440	610–1,294	578–931

National ambient air-quality standards for residential and industrial sites are 100 and $150 \mu\text{g m}^{-3}$, respectively

**Fig. 6** Assuming deposition fraction of 0.63 (Hind 1999; Chate 2011), diurnal variations in initial and deposited PM_{10} concentrations with sampling hours in Deepawali 2007 and 2008

Short-period averages of PM_{10} , $\text{PM}_{2.5}$, and PM_1 concentrations over Deepawali days were 723, 588, and $536 \mu\text{g m}^{-3}$ of 2007 and 501, 389, and $346 \mu\text{g m}^{-3}$ of 2008. Further, in general, highest PM_{10} , $\text{PM}_{2.5}$, and PM_1 levels were observed at 2000 h (on Deepawali day) and at 0500 h (on the next day). The mean CO, NO_2 , O_3 , and SO_2 concentrations were 2,400, 95, 40, and $19 \mu\text{g m}^{-3}$, respectively. Among gaseous pollutants, NO_2 and CO levels were frequently remained higher than the permissible limit of NAAQ (viz., 80 and $2,000 \mu\text{g m}^{-3}$ annual standards for NO_2 and CO). Short-term and long-term exposure to high levels of PM_{10} , $\text{PM}_{2.5}$, and PM_1 particles could cause health and environmental problems.

PM₁ and PM_{2.5} are hazardous to cardiovascular and respiratory syndromes in most polluted megacities as a result of dense vehicular traffic and extreme firework episodes in India (Deepawali festivals). Exceedance of PM₁ and PM_{2.5} beyond NAAQ standards during firework episodes is exacerbating health problem in Delhi. Rough estimates of inhaled particles show 37% deposition of total PM₁ within respiratory system on Deepawali and the day after.

To sum up, maximum pollution levels occur during winter while minimum during monsoon. Winter, therefore, has greater exposure risk as pollutants often get trapped in lower layers of the atmosphere thereby leading to higher pollution levels.

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